Keshav Rajput

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Software engineer, new grad. MS Computer Science, NYU Tandon, May 2026. C, C++, Python, TypeScript, Kotlin.
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EDUCATION

New York University, Tandon School of Engineering Sep 2024 - May 2026
Master of Science (MS) in Computer Science GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi 2020 - 2024
Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics CGPA 8.19/10

Coursework: Data Structures and Algorithms, Algorithms II, Database Systems, Distributed Systems, Operating Systems, Professional Software Development

PROJECTS

Atlas (in progress) | C++23, SDL_GPU, HLSL | github.com/happyc0der/atlas-engine 2026
- Built a C++23 engine for map-based strategy games solo across 32 gated milestones: SDL_GPU renderer, hot-reloading assets, versioned saves, 25 written design decisions.
- Made the simulation deterministic (fixed ticks, state hashing), so replays and two machines in lockstep over ENet stay bit-identical; a million-cell tick runs in 694 us.
- Sandboxed mods as WebAssembly with no clock, filesystem or randomness; over 1,000 tests, CI on macOS, Linux and Windows with ASan/TSan and benchmark gates.

mldsa-auth | C11, liboqs, libsodium, SQLite, CMake | github.com/happyc0der/mldsa-auth 2026
- Built a post-quantum mutual authentication protocol in C11: ML-DSA-65 (FIPS 204) signatures over hybrid X25519 + ML-KEM-768, HKDF-SHA256, ChaCha20-Poly1305 records.
- Measured a 0.45 ms median handshake and 736 MiB/s record encryption; ASan/UBSan CI on Linux and macOS, 12 libFuzzer targets, nightly mutation testing (299 must-kill mutations).
- Built mldsa-authd, a login service on it: login codes, device enrolment, revocation, key rotation, recovery codes; ProVerif model of the handshake.

doom-ports | C, eZ80 assembly, Monkey C, Connect IQ SDK | github.com/happyc0der/doom-ports 2026
- Wrote a first-person raycaster from scratch for the TI-84 Plus CE calculator in C and eZ80 assembly: 24 KB, 5 to 7 fps, all art generated at startup.
- Profiled the display (122 eZ80 cycles per VRAM byte), fit a frame-time model with a 6.2 fps ceiling, and moved the column fill to hand-written assembly.
- Ported it to the Garmin Venu X1 watch in Monkey C, rebuilt around a 0.2 ms per-draw cost: 1 to 2 fps to 15.6 fps; published on the Garmin Connect IQ store.

TideFit | TypeScript, Next.js 16, React 19, Supabase | github.com/UtkarshMitta/tide-fit 2026
App by Utkarsh Mittal; the audit, upgrade, tests, CI and launch are mine.
- Audited all 45 source files of a Next.js trip planner: 15 vulnerabilities confirmed on the running app (OAuth callbacks accepting any authorization code, a row-level security policy exposing every user), 13 fixed.
- Upgraded Next 14 to 16 and React 18 to 19 across 16 merged pull requests; npm audit 6 vulnerabilities to 0, tests 0 to 45, CI (typecheck, lint, tests) deploying to Vercel.
- Launched it to production at tide-fit.vercel.app: private file storage, an edge rate limit verified at 20 requests then 429, Strava and Google sign-in verified live.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Final rank 591 of 3,947 teams (top 15%); private leaderboard 0.918, public 0.952.
- **Gemma 4 Developer Agent, Kaggle (Google DeepMind)**, 2026. LLM agent entry in Google ADK; local evaluation harness with Docker sandboxes and tool calls, served by vLLM. In progress, ends 2026-12-02.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Systems: CMake, gRPC, PostgreSQL, SQLite, WebAssembly (sandboxed mods), Android (Jetpack Compose, Room), Next.js, React, FastAPI, Node.js, REST APIs, MCP

LLMs: LLM agents (Google ADK, MCP, tool calling), vLLM, Ollama, LiteLLM, LoRA/QLoRA fine-tuning (PEFT), llama.cpp/GGUF quantization

Testing: pytest, Catch2, vitest, Robolectric, libFuzzer, ASan/UBSan, static analysis (clang-tidy, ruff, ESLint), ProVerif, unit and integration tests, CI

Tools: Git, CI/CD (GitHub Actions), Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale, Claude Code

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes with regression tests to hawk-py
- Treasurer, OWASP Student Chapter, IIIT Delhi